# Project Report: Automatic Windshield Wiper System

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## 1 System Identification

The chosen embedded system is an **Automatic Windshield Wiper Controller** based on the Arduino Uno platform. This system autonomously manages vehicle visibility by detecting moisture levels via a resistive rain sensor and modulating the sweep frequency of a servo-driven wiper arm in real-time.

## 2 Real-World Operating Characteristics and Constraints

The system is designed to function within the following automotive-grade constraints:

- **Dynamic Latency:** The system maintains a tight feedback loop to ensure road safety. In heavy rain conditions, the response cycle is optimized to 500ms total (250ms per sweep direction) for maximum water displacement.
- **Operational Robustness:** The system utilizes threshold-based logic to handle varying water conductivity. It must reliably differentiate between a dry sensor ($> 900$ ADC units) and saturated conditions ($< 500$ ADC units) to avoid mechanical jitter.
- **Mechanical Constraints:** The sweep is software-limited to an arc between $45^\circ$ and $135^\circ$ to prevent the wiper from overextending beyond the windshield frame.

## 3 Real-World Interfaces

The system utilizes specific hardware interfaces to bridge software logic with physical phenomena:

- **Moisture Transduction Interface:** A rain sensor module (MH-RD) translates the physical presence of water into a variable analog voltage signal.
- **Angular Actuation Interface:** A micro servo (SG90) converts Pulse Width Modulation (PWM) signals into precise physical movement.
- **Diagnostic Interface:** A Serial UART interface (9600 baud) provides a live data stream of environmental values for system calibration.

## 4 Architectural IPs and Justification

The following internal microcontroller Intellectual Property (IP) blocks are utilized:

| Interface | Architectural IP | Justification |
|---|---|---|
| Rain Sensor | Analog-to-Digital (ADC) | Uses the internal 10-bit ADC (Pin A0) to convert sensor voltage into 1024 discrete digital levels. |
| Servo Motor | Timer/PWM Generator | Employs 16-bit hardware timers to generate a precise $50Hz$ duty cycle for position control. |
| Logic Timing | CPU / Clock Tree | Utilizes the instruction clock for `delay()` functions to regulate wiper speed based on rainfall intensity. |
| Communication | UART (Universal Serial) | Enables asynchronous data transmission to the PC for real-time monitoring of the `rainValue`. |

## 5 System Boundaries and Logic Definition

The system logic is bounded by four distinct operational states based on the Analog Input (A0) readings:

1. **Idle State ($>$ 900):** No moisture detected. The wiper remains at a fixed 90° "home" position.
2. **Low Intensity ($700 - 900$):** Slowest sweep speed with an 800ms delay between 70° and 110° positions.
3. **Medium Intensity ($500 - 700$):** Moderate sweep speed with a 500ms delay and an expanded 60° to 120° arc.
4. **High Intensity ($<$ 500):** Maximum frequency with a 250ms delay and the widest 45° to 135° clearing range.